

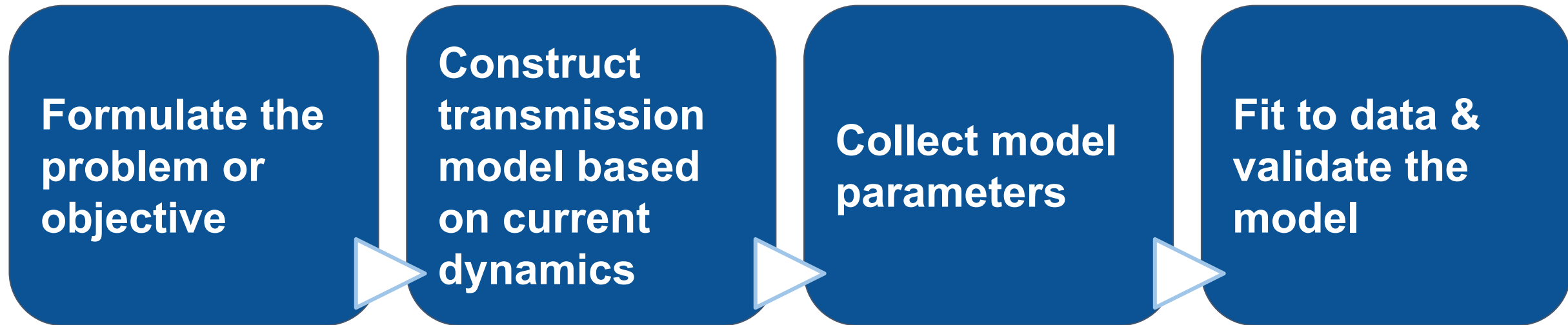
Heterogeneity and Age Patterns

Recap

Day 1

RECAP

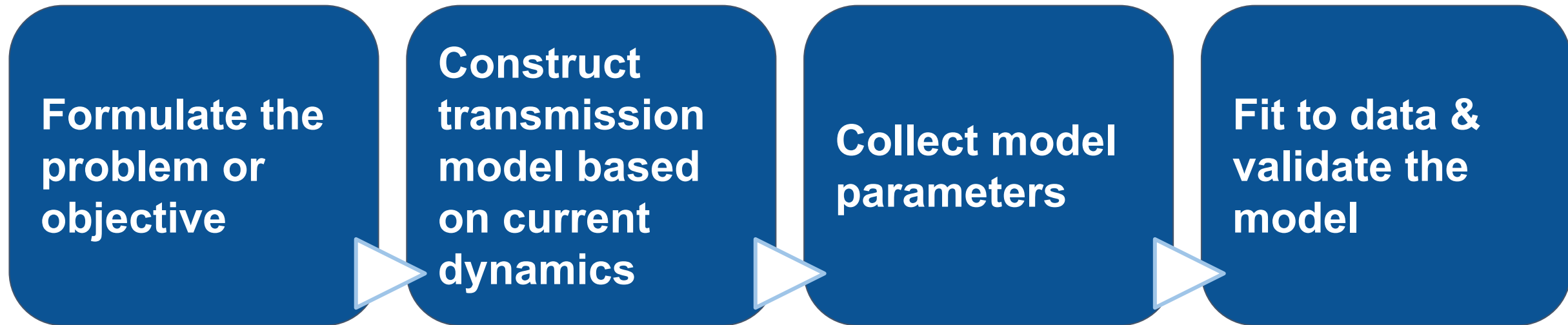
Steps of Developing a Model



- Objective informs model utility (*theoretical, inference, strategic, forecast*) informs model type

RECAP

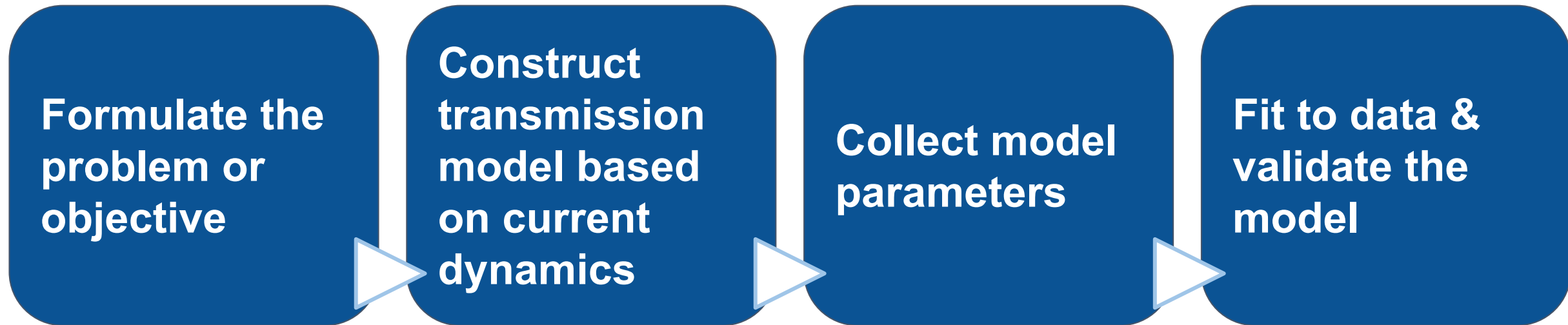
Steps of Developing a Model



- The basic SIR model
- Extension of the basic model:
vaccination activities

RECAP

Steps of Developing a Model



- How data are related to models
incidence, prevalence, seroprevalence
- Critical model parameters:
 R_0 and its components, β and γ

Review of basic compartmental models

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

- **S**, **I**, and **R** represent the number of individuals currently **susceptible**, **infected**, and **recovered**
- Contact process quantifies the rate at which **susceptibles** and **infecteds** interact
- Transmission parameter: β is rate of infectious contact * probability of infection given contact. High β = more transmission
- Higher recovery rate γ = shorter duration of infection

Heterogeneity and Age Patterns

Learning Objectives

By the end of this session you should learn:

- How to define and estimate force of infection
- The utility of heterogeneity in models
- The specific utility of age heterogeneity in models
- The relationship between mean age of infection to FOI and R_0
- The difference between age contact patterns, WAIFW, and FOI

Force of infection (λ)

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

$$\lambda = \beta I$$

per capita rate at which susceptible individuals contract the infection OR transmission rate per susceptible individual

$\therefore \beta SI$ or $S\lambda$ is the total transmission rate of the entire susceptible population

βSI assumes **homogeneous mixing** in the population which means everyone interacts with equal probability with everyone else

Modeling Heterogeneity

using structured models

Host Heterogeneities

- The **basic SIR** model only compartmentalizes the populations by infection status and history (**one degree of subdivision**).
- In this lecture we introduce **second degree of subdivision** in which all individuals in each second degree subdivision will have the same parameters (e.g., β).
- By taking into account heterogeneities of the second degree of subdivision, we create a **structured model**.

Foot and Mouth Disease Example

Foot-and-mouth disease (FMD)

- Caused by the FMD virus (FMDV)
- Affects multiple species of mammals (including cows, sheep, goats, pigs)
- Can cause blisters around the hooves and mouth and spontaneous abortions
- Reduces milk yield and negative economic impact

Example – Foot-and-mouth (FMD)

Where does heterogeneity appear in the case of FMD?

- Pathogen: FMDV affects species differently
- Host: animals have different lifespans and behaviours
- Policy: some countries vaccinate cows, others also vaccinate pigs
- Environment: contact rates within and between species can differ based on how animals are kept

FMDV is different in 🐷 and 🐮 !

second degree subdivision = species



- FMDV is highly transmissible in pigs compared to other species
- Pigs tend to have a shorter lifespan in animal husbandry contexts (1 year or less)
- Pigs are not always vaccinated in FMD control programs



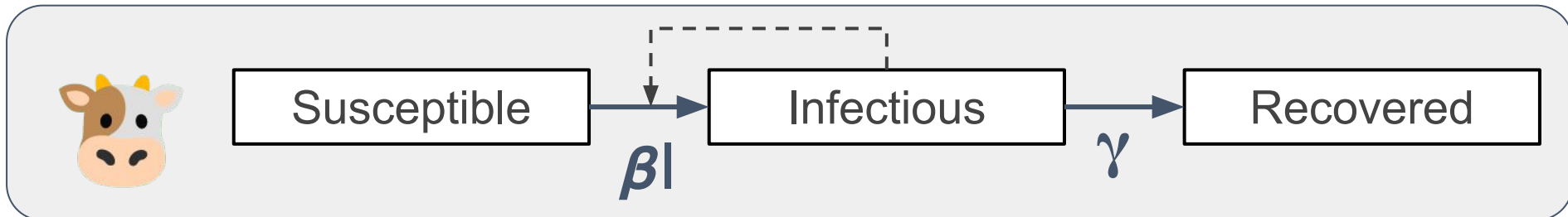
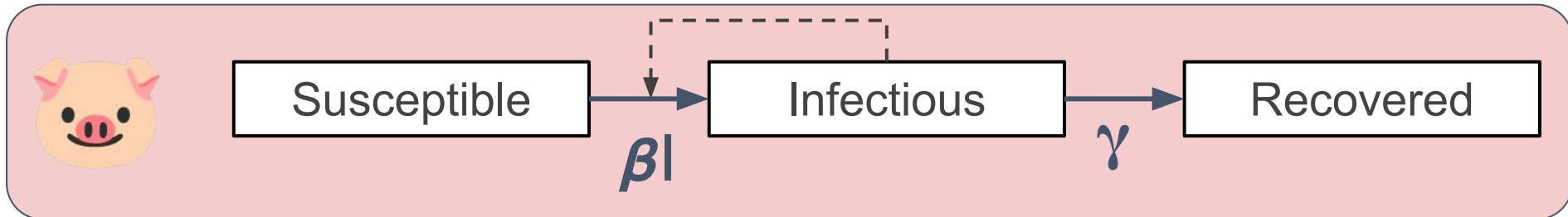
- Cattle are more susceptible to FMD compared to other species
- Cattle can live up to 14 years in some animal husbandry contexts (e.g., in India)
- Cattle are almost always vaccinated in FMD control programs

FMDV is different in 🐷 and 🐮!

How do we model those differences?

1. Make 2 separate models

Why is this a bad idea?

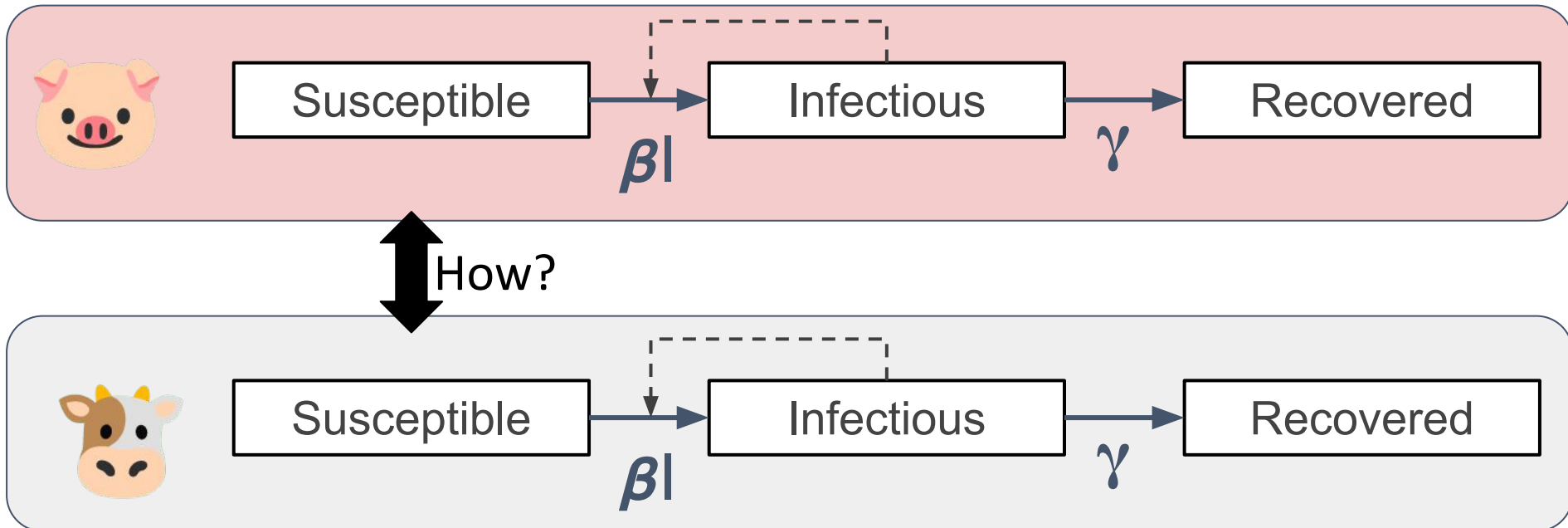


FMDV is different in 🐷 and 🐮!

How do we model those differences?

1. Make 2 separate models
2. Make one model with both species ✓

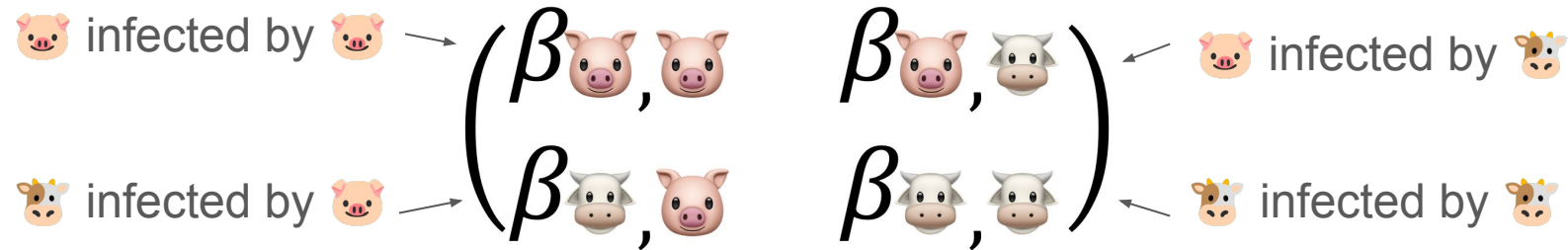
Why is this a bad idea?



One model - Two species

“Talking” to each other via transmission parameter

Instead of having a single β term for transmission (reminder: β is rate of infectious contact * probability of infection given contact), we use a $m \times m$ matrix of β values for m different species in our model.



This **transmission matrix**, **WAIFW** (Who Acquires Infection from Whom), captures the transmission between the two groups.

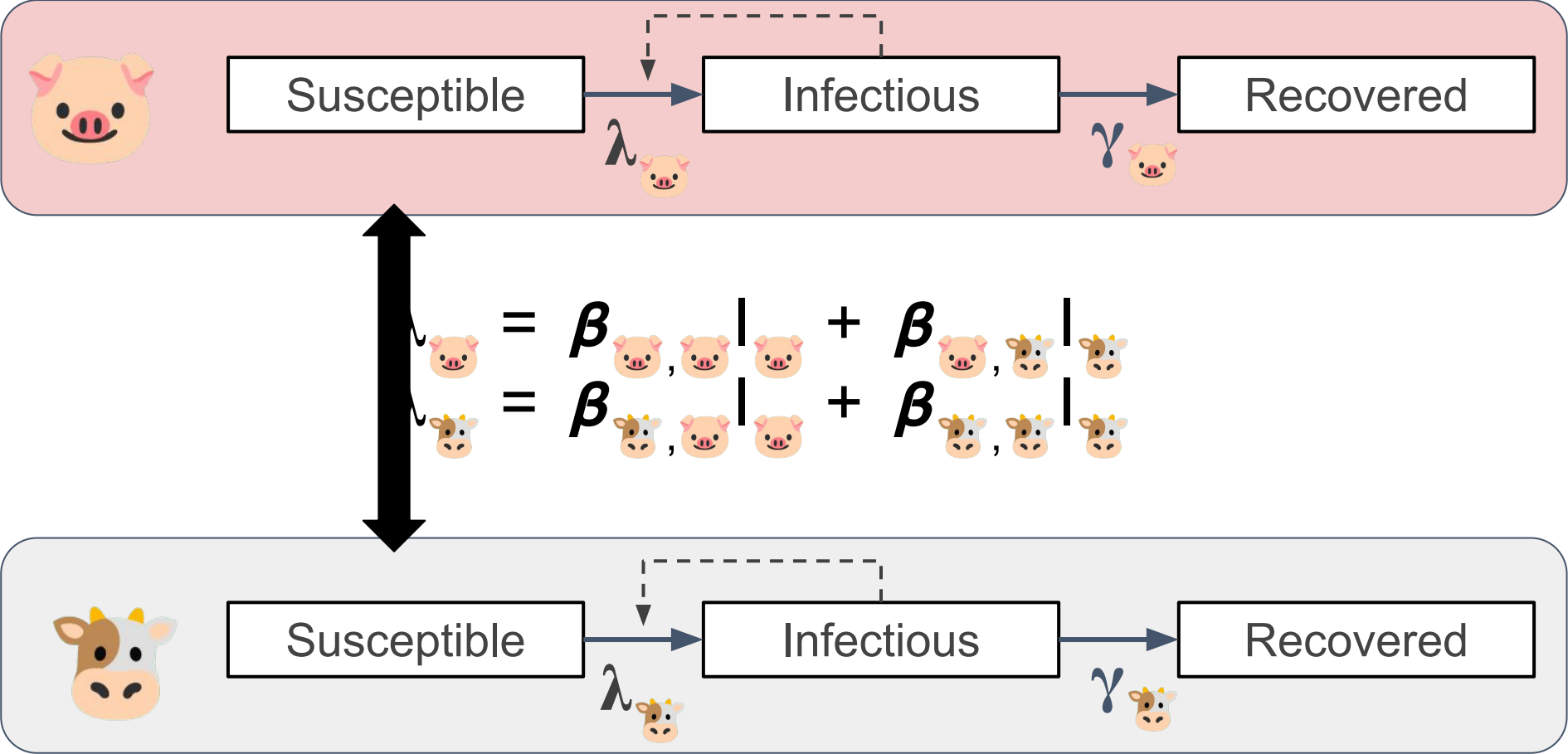
One model - Two species

“Talking” to each other via transmission parameter

As a result, the force of infection (reminder: $\lambda = \beta I$ is per capita rate at which susceptible individuals contract the infection) is based on the transmission matrix $m \times m$ of β values, as well as an I term for each m species.

$$\begin{aligned}\lambda_{\text{pig}} &= \beta_{\text{pig,pig}} I_{\text{pig}} + \beta_{\text{pig,cow}} I_{\text{cow}} \\ \lambda_{\text{cow}} &= \beta_{\text{cow,pig}} I_{\text{pig}} + \beta_{\text{cow,cow}} I_{\text{cow}}\end{aligned}$$

Species structured HFMD model



Host Heterogeneities

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- In this lecture we introduce **second degree of subdivision** in which all individuals in each second degree subdivision will have the same parameters (e.g., β).
- By taking into account heterogeneities of the second degree of subdivision, we create a **structured model**.

Host Heterogeneities

- **Advantages of modeling heterogeneities:** more accurate models; we can determine prevalence of infection in each second degree subdivision; we can assess more targeted and effective control measures
- **Disadvantages of modeling heterogeneities:** Incorporating heterogeneities increases the number of equations and its sequel (i.e, the number of parameters we need to collect/estimate)

What other heterogeneities
might be important to model?

Non-directional vs Directional Transitions

Heterogeneity with non-directional transitions:

- Risk groups in STI settings (e.g., using a condom vs. not using a condom)
- Vaccine status (e.g., COVID-19 vaccine impacts beta and gamma)
- Sex
- Number of contacts (e.g., superspreaders vs not superspreaders)
- Co-morbidities or immune suppression

Heterogeneity with directional transitions:

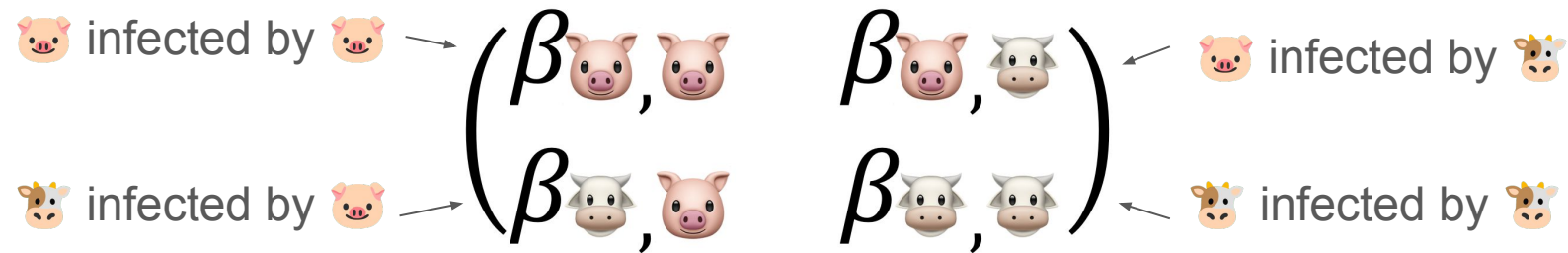
- Aging
- Life stages in animals
- Time since infection

Heterogeneity by Age

Directional Transitions

Modeling age heterogeneity

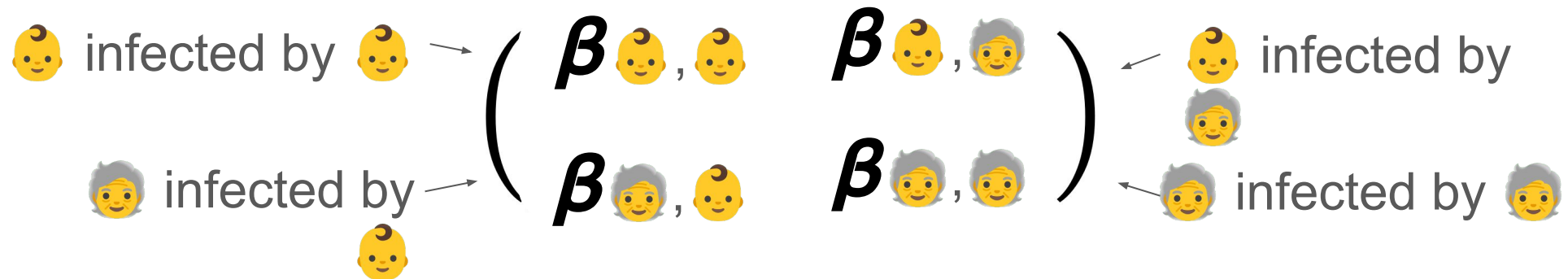
It is the **same as modeling heterogeneity with non-directional transitions** (i.e., age-specific parameters and a matrix of β or WAIFW to “talk” to each other), but



2 populations, therefore WAIFW 2x2 matrix (4 β s)
diagonal (top left to bottom right) - assortative transmission

Modeling age heterogeneity

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Modeling age heterogeneity

It is the **same as modeling heterogeneity with non-directional transitions** (i.e., age-specific parameters and a matrix of β or WAIFW to “talk” to each other), but

$$\begin{pmatrix} \beta_{\text{baby}, \text{baby}} & \beta_{\text{baby}, \text{adult}} \\ \beta_{\text{adult}, \text{baby}} & \beta_{\text{adult}, \text{adult}} \end{pmatrix}$$

WAIFW matrix

diagonal is assortative transmission

Modeling age heterogeneity

It is the same as modeling heterogeneity with non-directional transitions (i.e., age-specific parameters and a matrix of β or WAIFW to “talk” to each other), but with the addition of individuals ageing

$$\begin{pmatrix} \beta_{\text{baby}, \text{baby}} & \beta_{\text{baby}, \text{elderly}} \\ \beta_{\text{elderly}, \text{baby}} & \beta_{\text{elderly}, \text{elderly}} \end{pmatrix}$$

WAIFW matrix

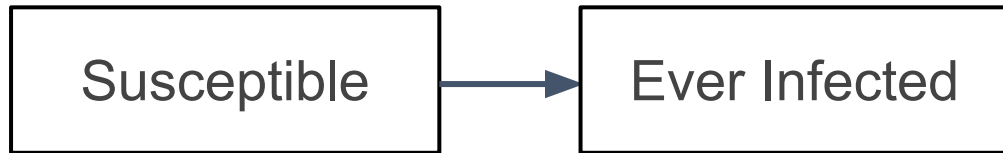
diagonal is assortative transmission

$$\begin{pmatrix} \text{baby} \\ \text{elderly} \end{pmatrix} \begin{pmatrix} -a & 0 \\ a & -\mu \end{pmatrix}$$

ageing matrix

from column age group to row age group

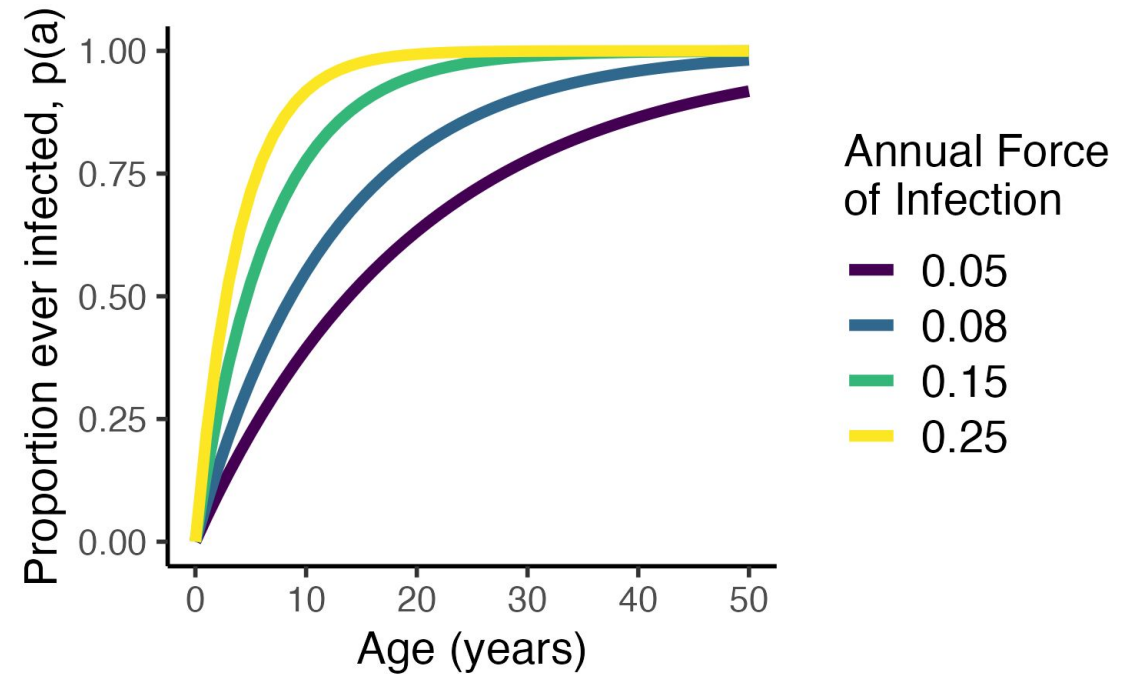
In childhood infections, ageing alone is associated with disease transmission



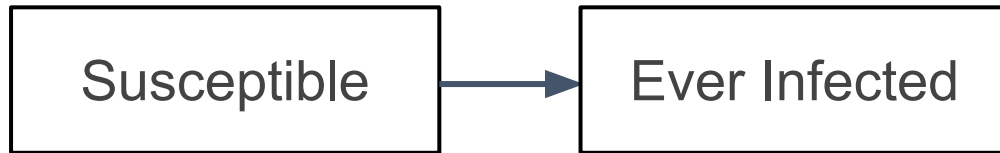
$$p(a) = 1 - e^{-\lambda a}$$

Simple Catalytic Model:

Assumes constant force of infection and homogeneously mixing population



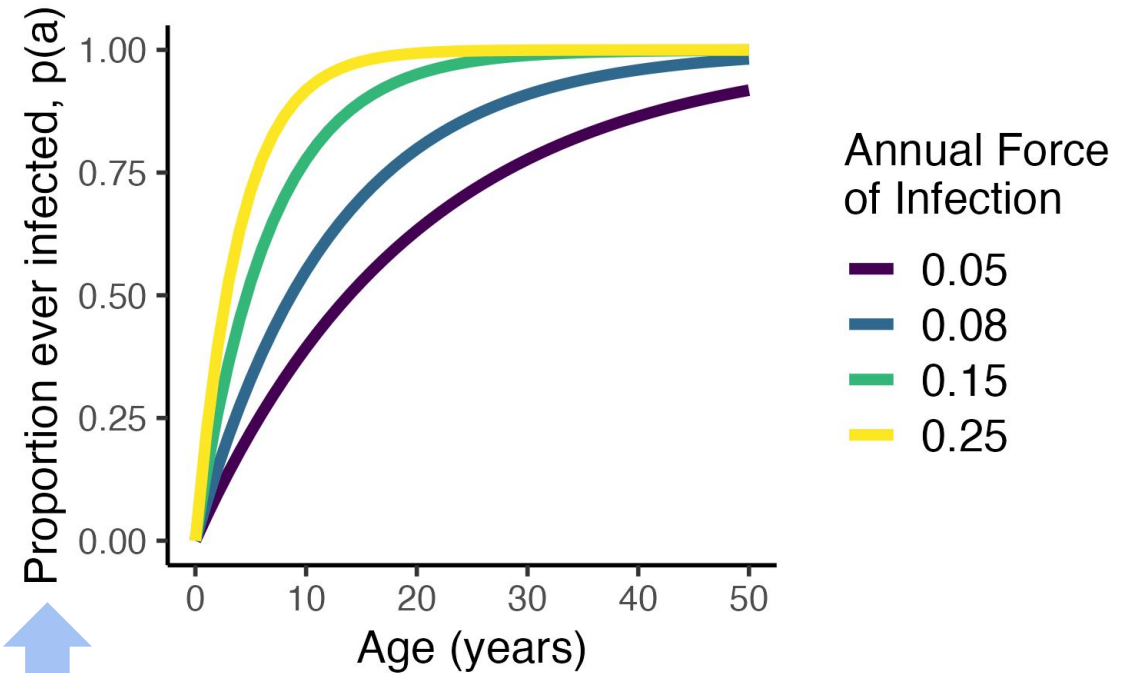
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Simple Catalytic Model:

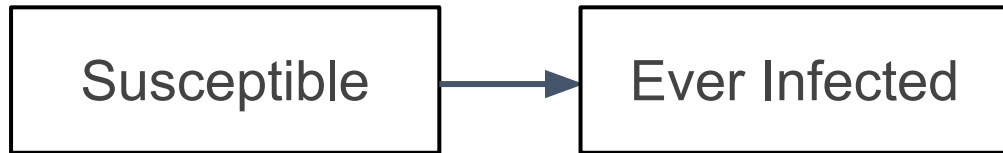
Assumes constant force of infection and homogeneously mixing population



- R class in SIR model
- Proportion with IgG antibody > protective level
- Proportion seropositive

- Proportion immune
- Seroprevalence

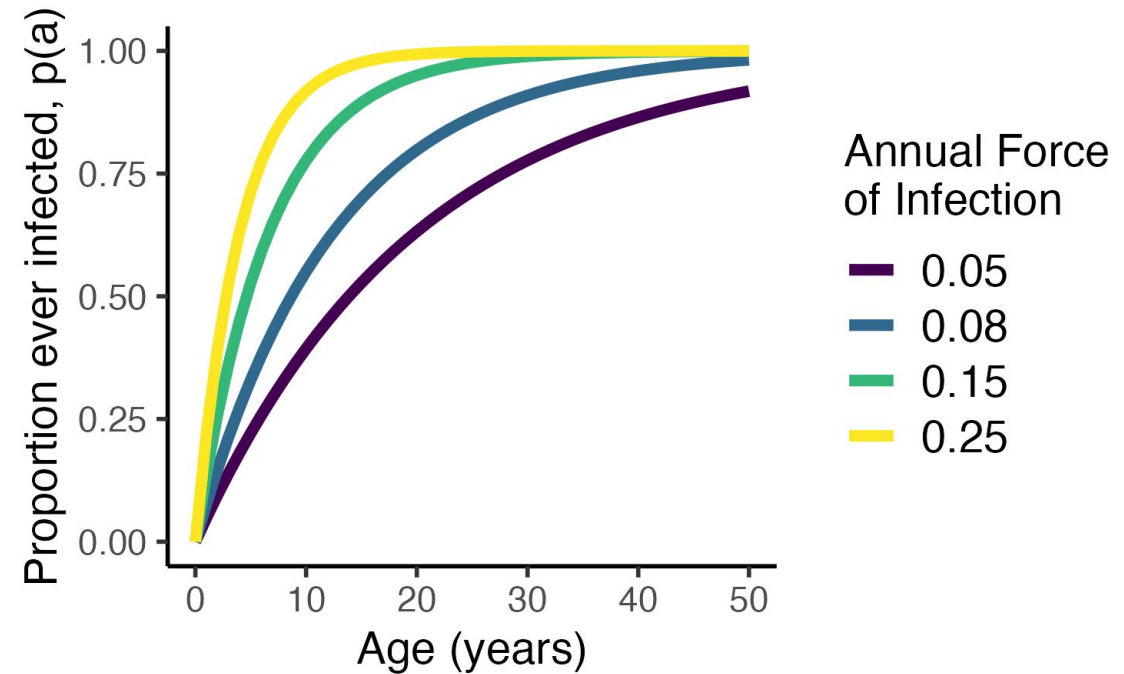
In childhood infections, ageing alone is associated with disease transmission



$$p(a) = 1 - e^{-\lambda a}$$

Simple Catalytic Model:

Assumes constant force of infection and homogeneously mixing population



Reminder: The average time to event = $1/\{\text{rate at which event occurs}\}$

Average age of infection, $A \approx 1/\lambda$

Mean age of infection and R_0

- The mean age is the average time from birth to infection
- Important indicator of prevalence

$$A \approx 1/\lambda$$

some substitutions given SIR model with births/deaths (μ) $\rightarrow$

$$A \approx 1 / \mu(R_0 - 1)$$

$$A \approx L / (R_0 - 1) \text{ where } L \text{ is life expectancy}$$

Reminder:

$$\begin{aligned}\frac{dS}{dt} &= \mu N - \beta SI - \mu S \\ \frac{dI}{dt} &= \beta SI - \gamma I - \mu I \\ \frac{dR}{dt} &= \gamma I - \mu R\end{aligned}$$

- This basic calculation requires strong assumption that age-specific force of infection is constant

Intuition of mean age of infection and R_0

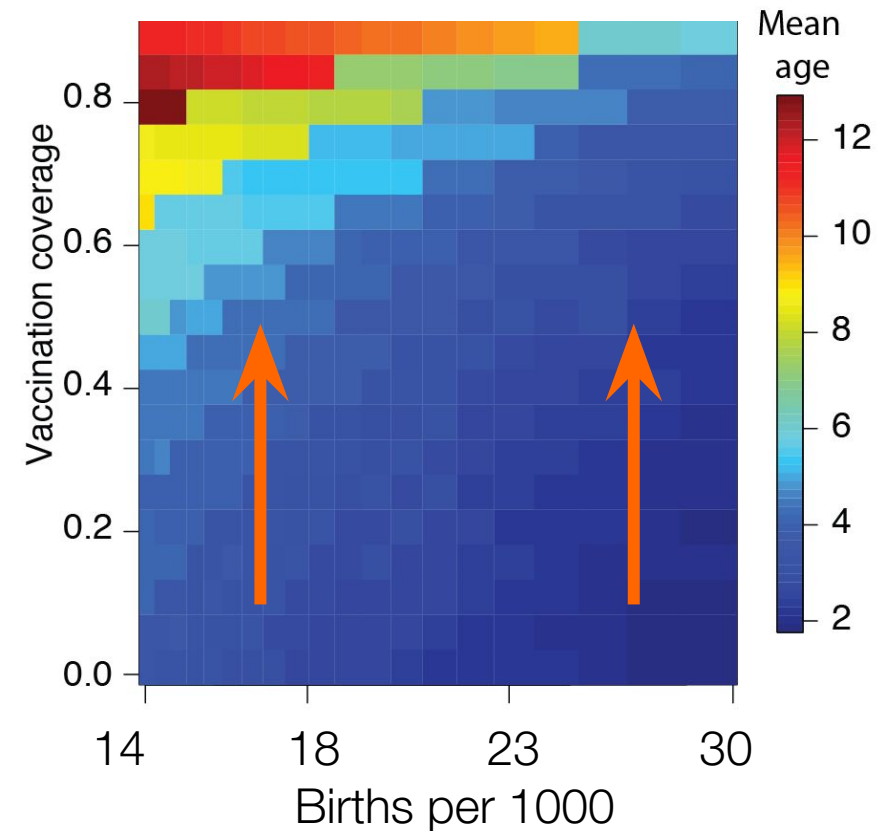
$$A \approx 1 / \mu(R_0 - 1)$$

- Reduced R_0 results in lower force of infection on each susceptible individual
- Meaning a longer wait until contact between susceptible and infectious individuals
- Resulting in higher mean age at infection

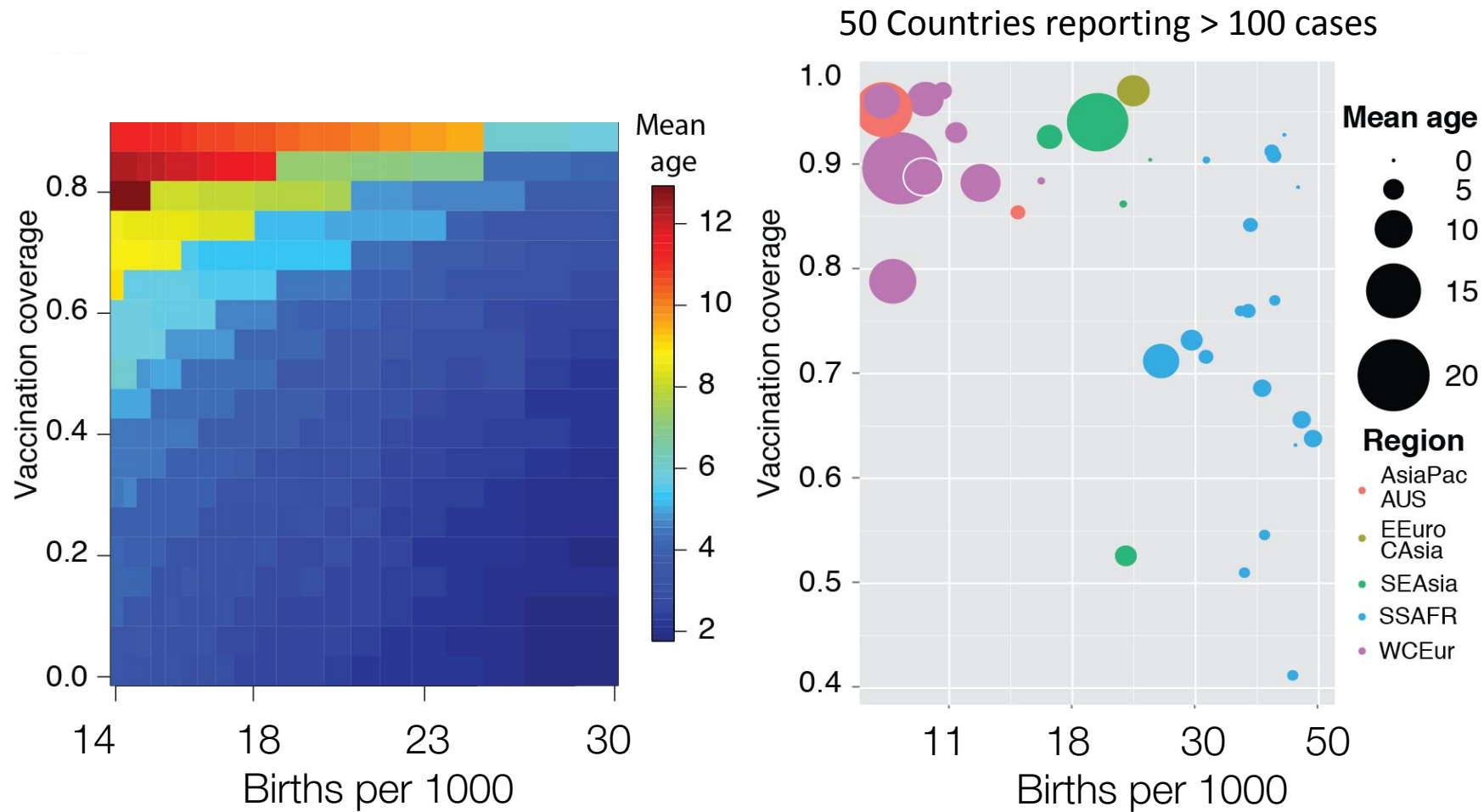
Intuition of mean age of infection and R_0

$$A \approx 1 / \mu(R_0 - 1)$$

- Reduced R_0 results in lower force of infection on each susceptible individual
- Meaning a longer wait until contact between susceptible and infectious individuals
- Resulting in higher mean age at infection
- The absolute value of mean age is mediated by underlying demographic rates



Observation About Mean Age of Infection



Estimating Force of Infection

J. Hyg., Camb. (1985), **95**, 419–436
Printed in Great Britain

419

The estimation of age-related rates of infection from case notifications and serological data

BY B. T. GRENFELL AND R. M. ANDERSON

*Department of Pure and Applied Biology, Imperial College, London University,
London SW7 2BB*

Empirical evidence based on:

- age-specific serological data OR
- age distribution of infected individuals

Why age-specific FOI?

24

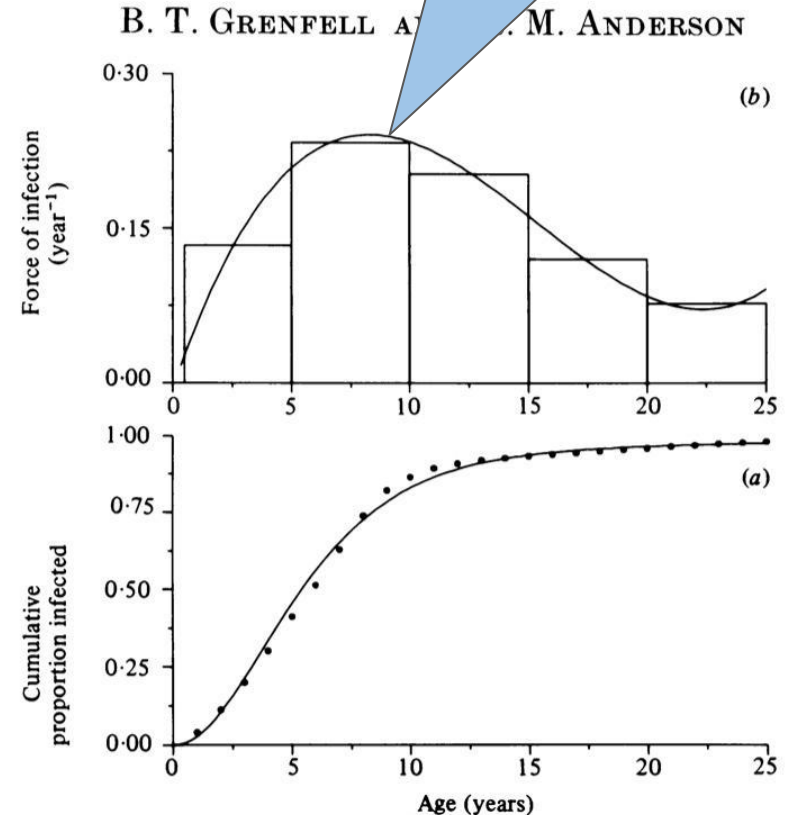
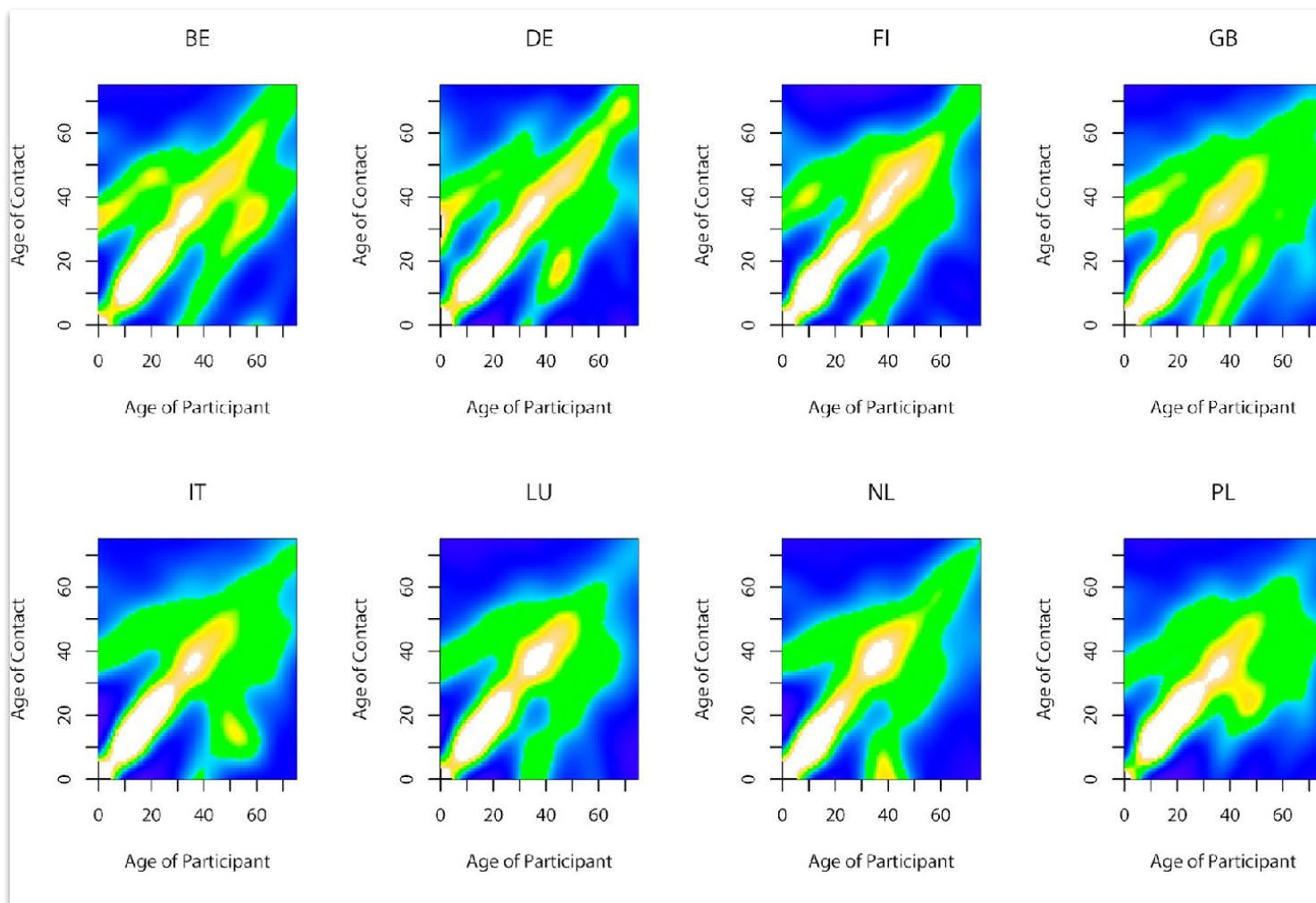


Fig. 1. Analysis of measles notifications for Baltimore, USA, 1906–15. (a) Observed and expected cumulative proportions infected by age ($F(a)$) in equation (5). (b) The fitted force of infection polynomial ($\lambda(a)$) in equation (4). Here and in Figs. 2–7 the histogram represents average force of infection estimates (in the age ranges 0–5–5 years, 5–10 years, etc.) derived from the fitted polynomial, which is documented in Table 1.

Age-specific contact patterns

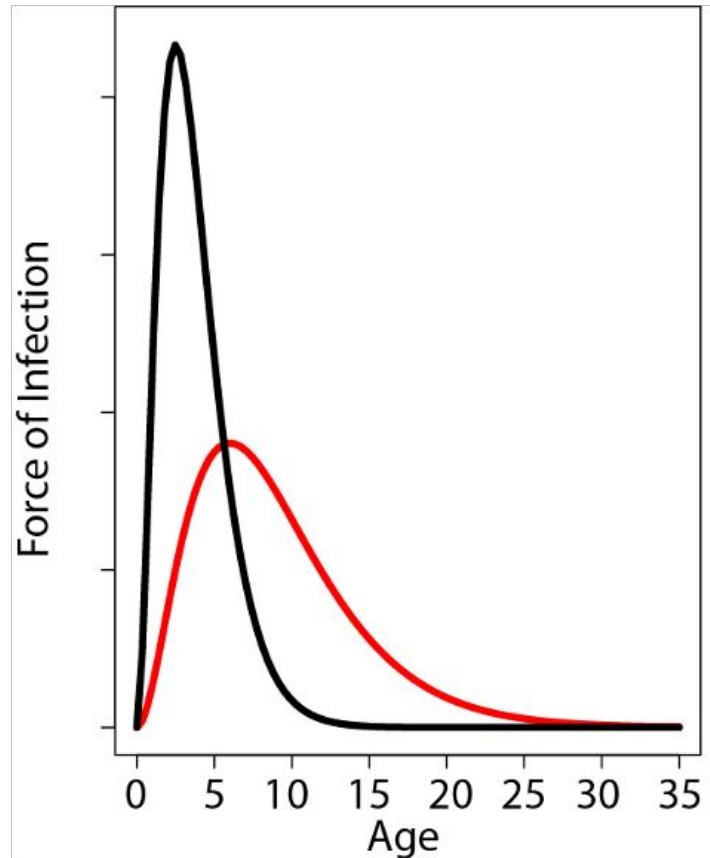


“age assortative contacts” =
lots contacts along diagonal

How does this differ from
WAIFW?

How does this differ from
age-specific FOI?

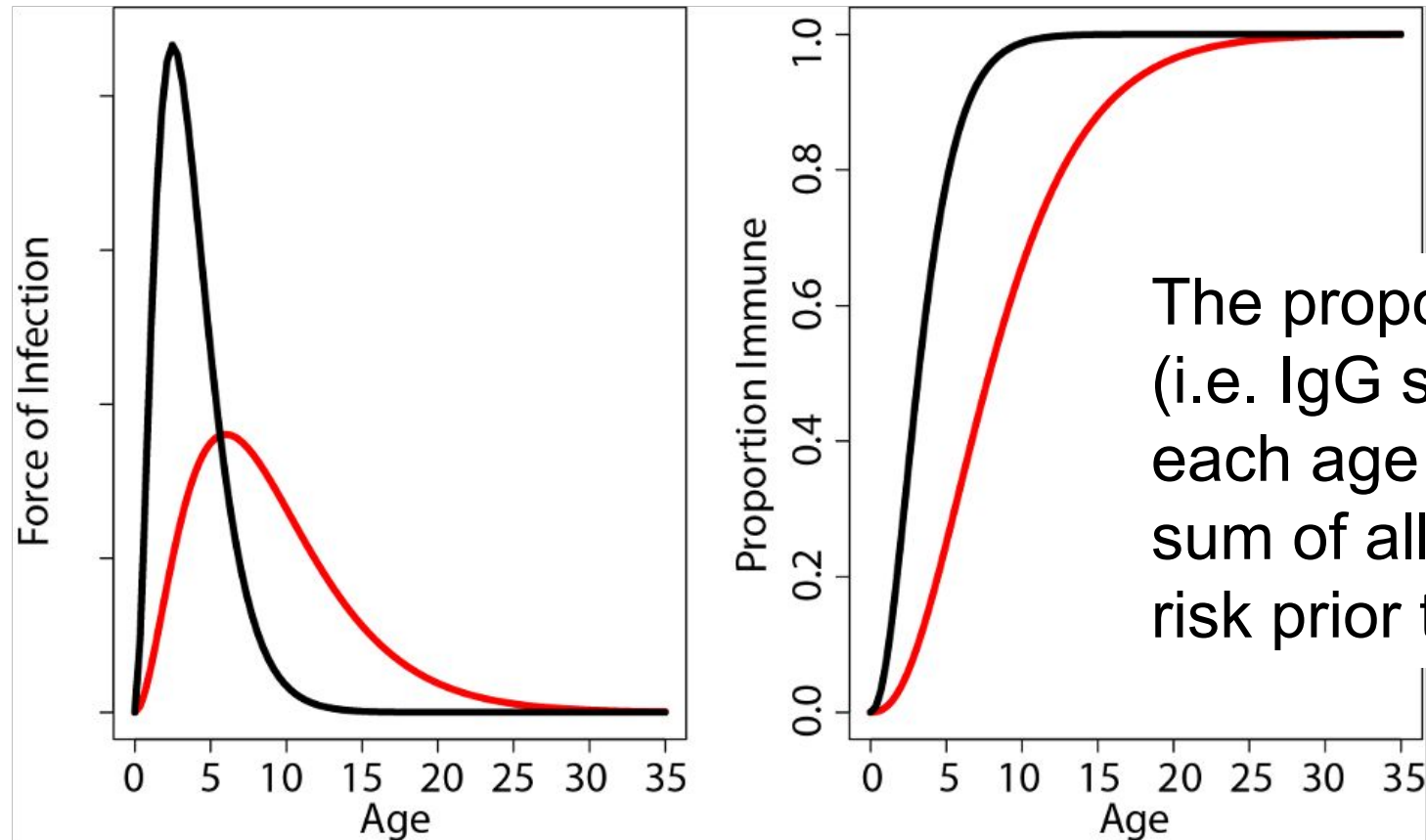
Age-Specific FOI



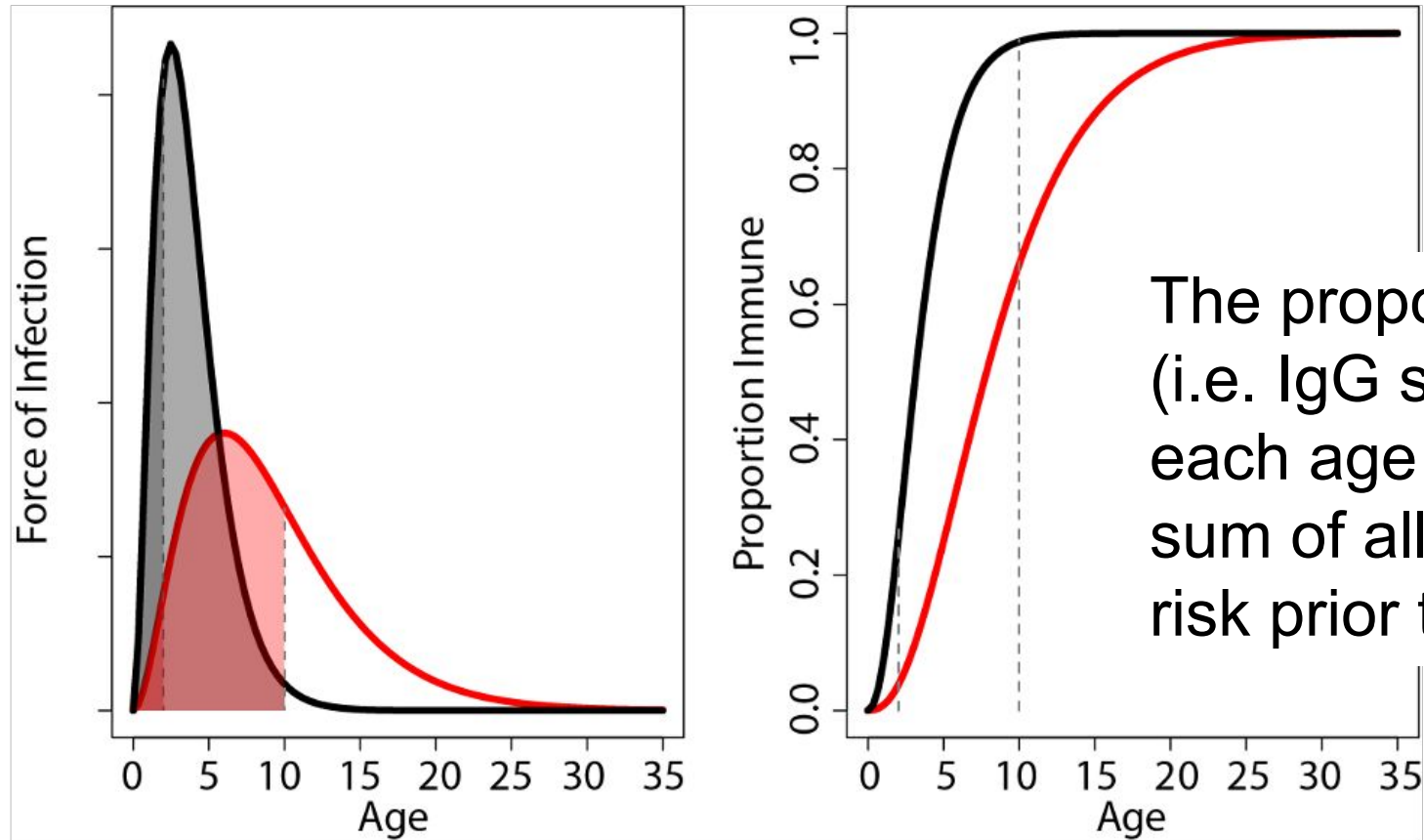
The age specific Force of Infection is the rate at which individuals of each age are exposed to infection

The shape of this function reflects the absolute risk (height) and the age ranges over which infection is most likely (breadth of the curve)

Age-Specific FOI



Age-Specific FOI



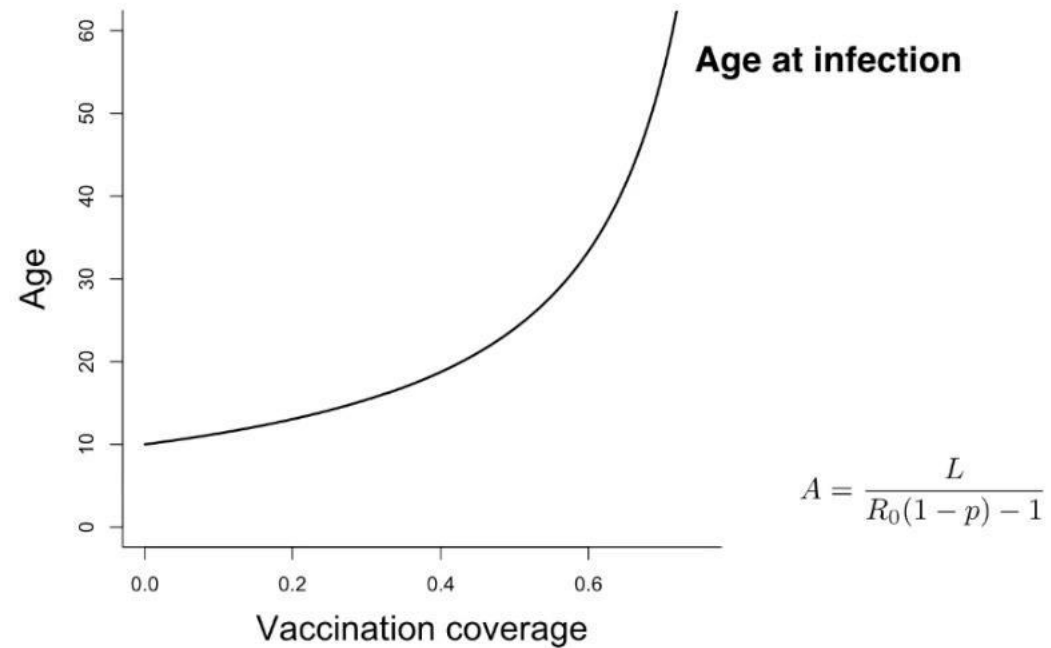
Age-specific severity

- Most severe in the young:
 - Measles, pertussis, diphtheria
- Most severe in the elderly
 - COVID-19
- Most severe in the young AND the elderly
 - Influenza
- Most severe in intermediate ages
 - Zika virus, rubella – severe complications in pregnancy

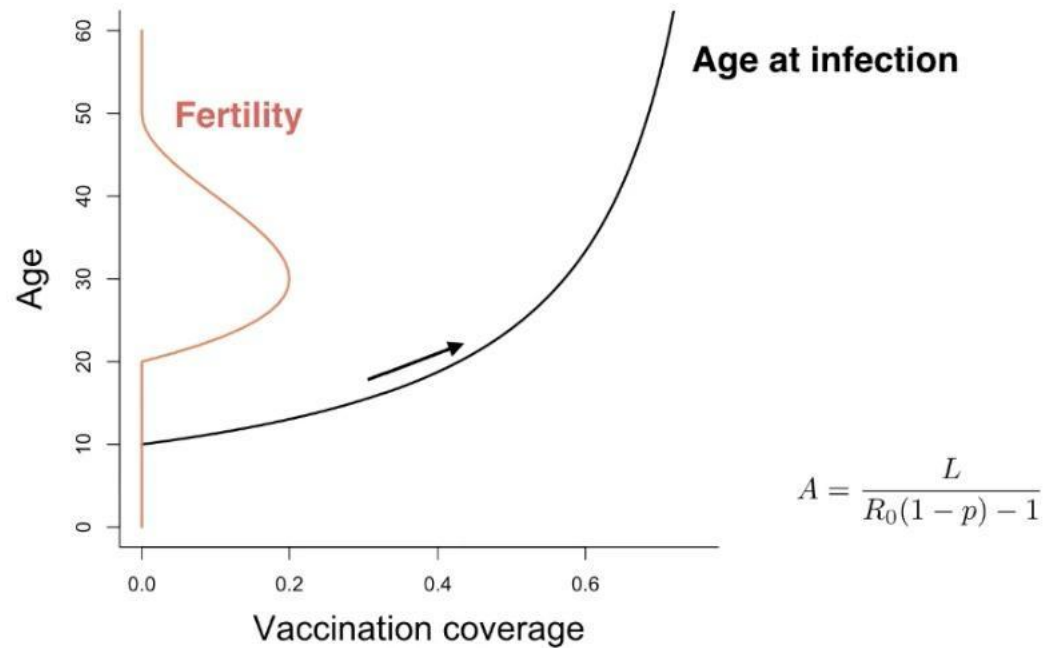
Rubella and CRS

- Rubella is a directly transmitted virus with R_0 of 2-6 in endemic regions
- Infections in children and adults are mild
- Infections during first trimester of pregnancy can lead to serious complications (Congenital Rubella Syndrome, CRS)
 - Deafness
 - Blindness
 - Congenital heart disease

Age Dynamics Following RCV Introduction



As vaccination increases average age increases moving into fertility ages

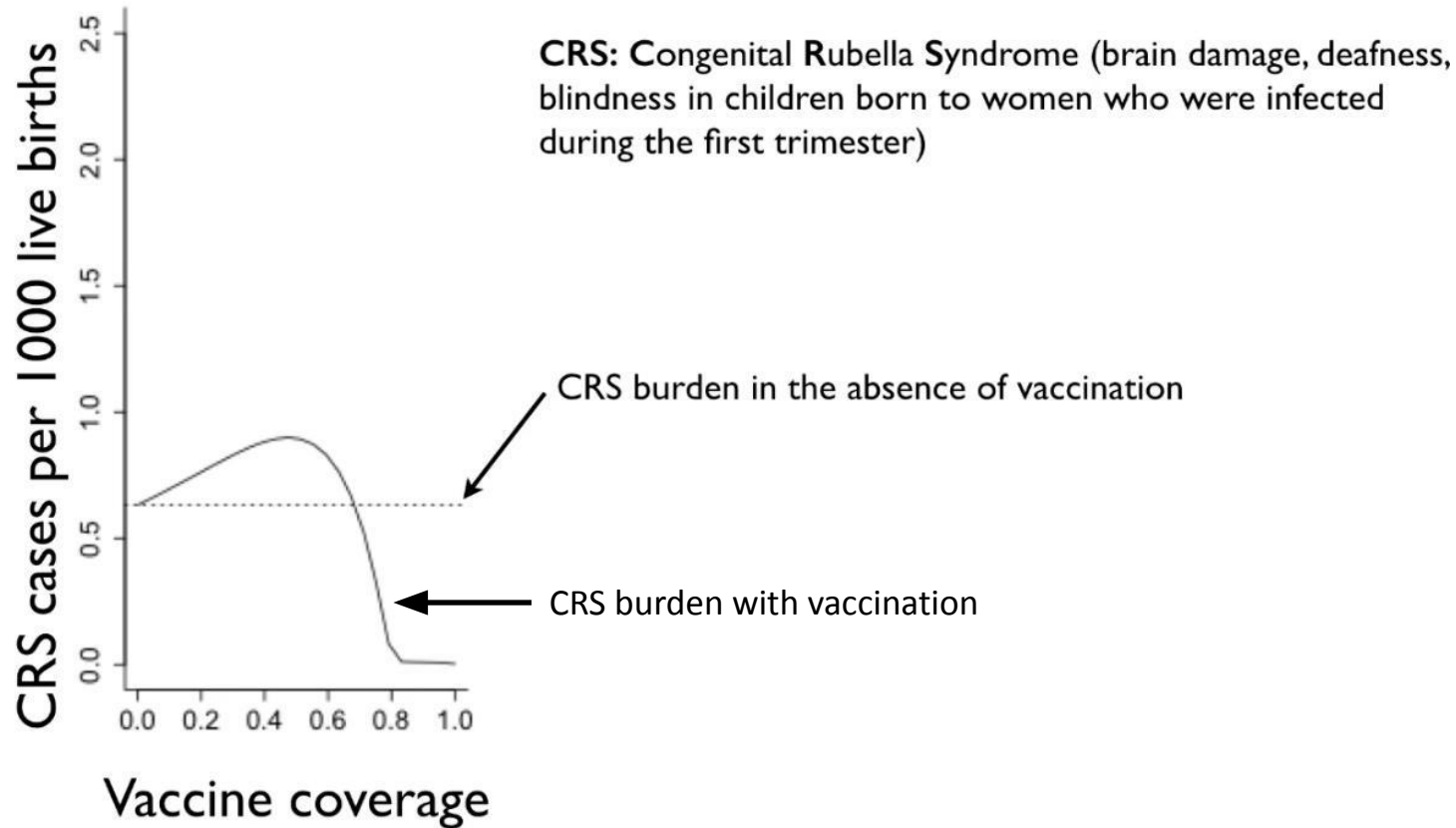


Because increase vaccination increases average age of infection, it also impacts the number of CRS cases

Epidemiol. Infect. (2012), **140**, 2290–2301. © Cambridge University Press 2012
doi:10.1017/S0950268812000131

**Impact of birth rate, seasonality and transmission rate
on minimum levels of coverage needed for rubella vaccination**

C. J. E. METCALF^{1,2*}, J. LESSLER³, P. KLEPAC³, F. CUTTS⁴
AND B. T. GRENFELL^{2,5}

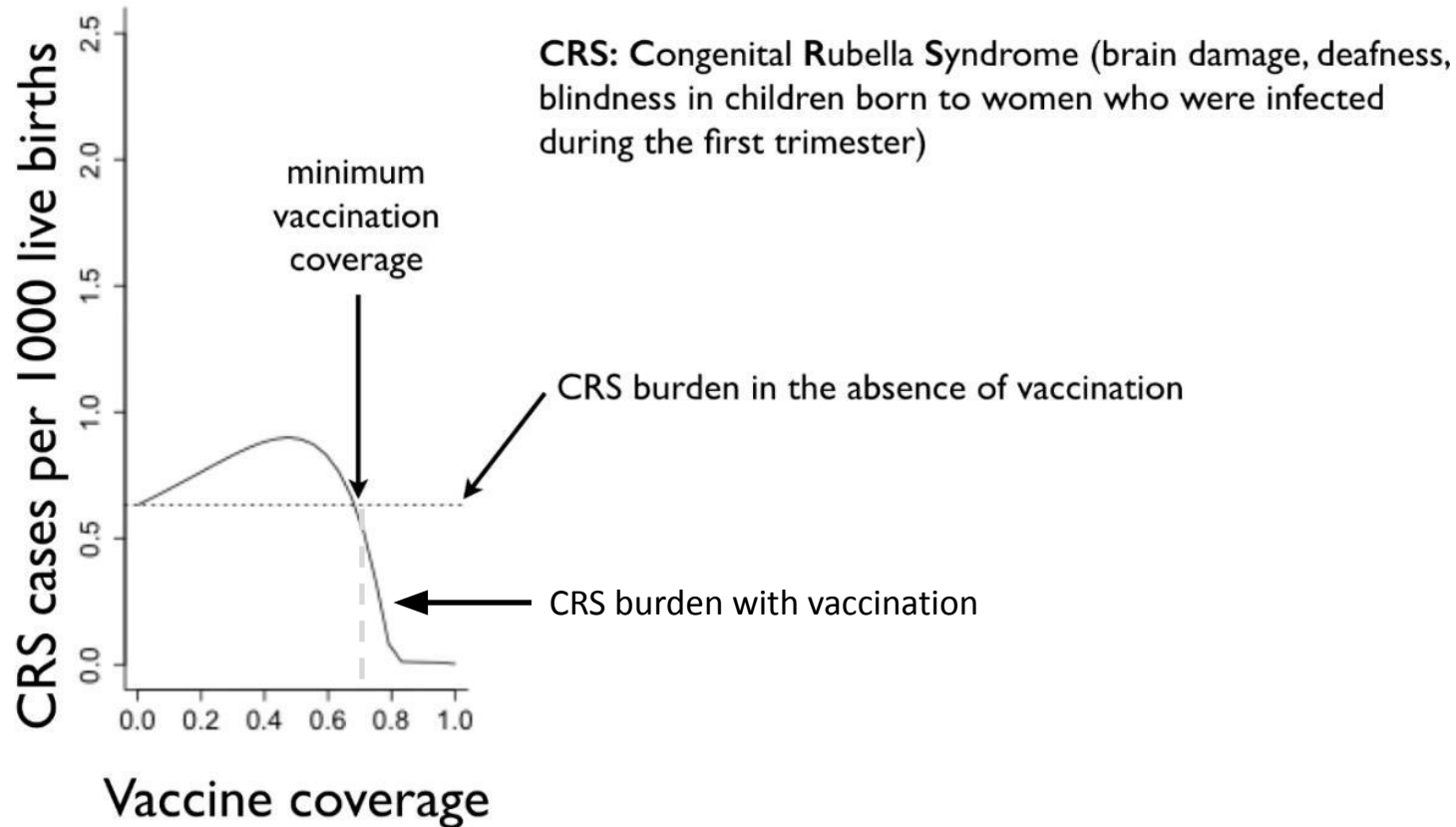


To avoid vaccination increasing the incidence of CRS, there is a minimum vaccination threshold

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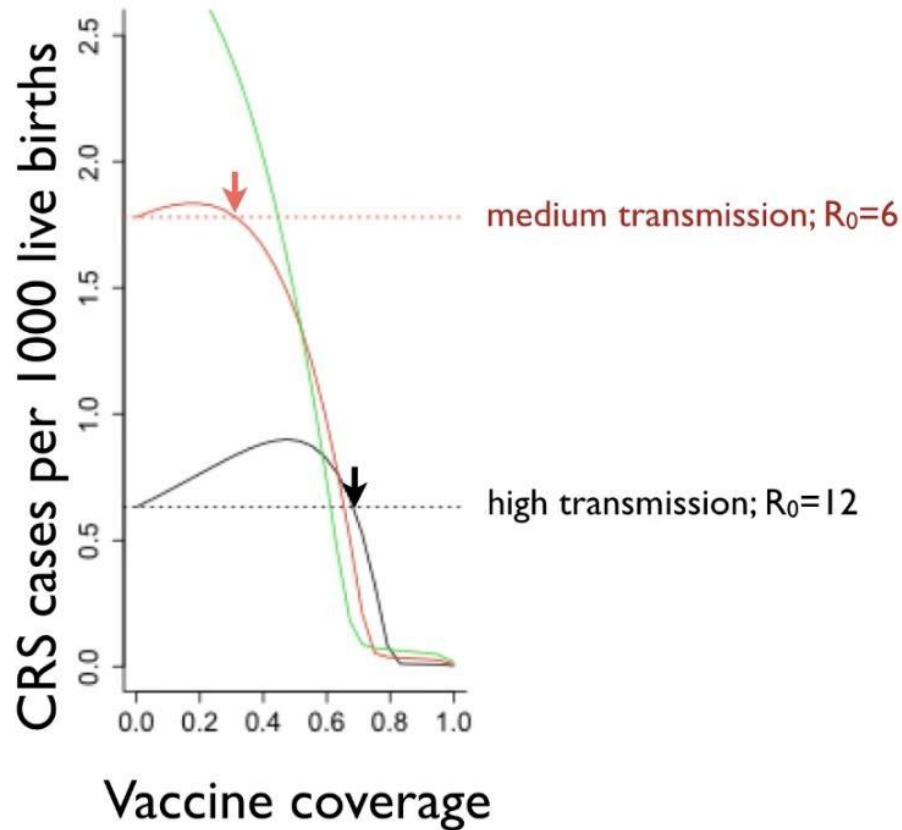


The minimum vaccination threshold depends on the R_0 of rubella

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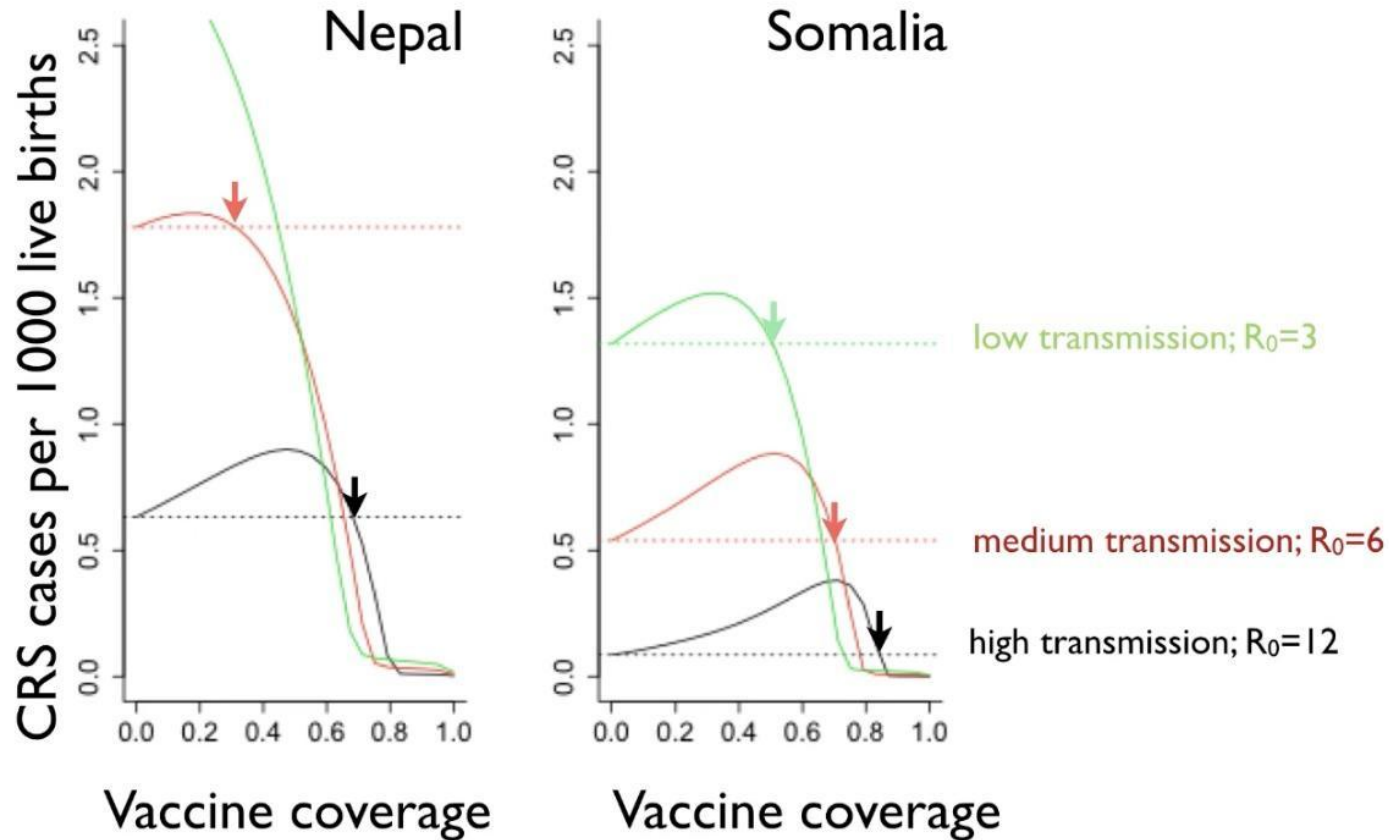


The minimum vaccination threshold also depends on the population demography

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RCV Policy

- Since 2000, there was a single WHO policy recommendation that countries needed to reach 80% of birth cohorts with vaccination in order to introduce RCV
- As of September 2024, that has been changed (using modeling) to show that 80% was too conservative
 - The optimal strategy for introduction is highly dependent on local vaccination coverage (routine plus campaigns), local transmission rate (lower R_0 than modeled prior), and local demography (birth rates declining)

Summary: What age heterogeneities matter for IDs?

- Age-specific **contacts** → Age-specific **transmission**
- Age-specific **disease progression / outcomes**
- Age-specific **infectiousness** (e.g., viral shedding by age)
- Age-specific **susceptibility** (e.g., maternally derived immunity)
- Age-dependent **vaccination effectiveness**
- Age-specific **prevention** strategies
- Age-specific **treatments**
- Age-specific **surveillance** bias

Modeling Age Heterogeneity

Modeling age heterogeneity

It is the **same as modeling heterogeneity with non-directional transitions** (i.e., age-specific parameters and a matrix of β or WAIFW to “talk” to each other), but **with the addition of individuals ageing**

$$\begin{pmatrix} \beta_{\text{BB}}, \beta_{\text{BG}} & \beta_{\text{GB}}, \beta_{\text{GG}} \\ \beta_{\text{BB}}, \beta_{\text{BG}} & \beta_{\text{GB}}, \beta_{\text{GG}} \end{pmatrix}$$

WAIFW matrix

diagonal is assortative transmission

$$\begin{pmatrix} \text{BB} \\ \text{BG} \end{pmatrix} \begin{pmatrix} -a & 0 \\ a & -\mu \end{pmatrix} \begin{matrix} \text{BB} \\ \text{BG} \end{matrix}$$

deaths

ageing matrix

from column age group to row age group

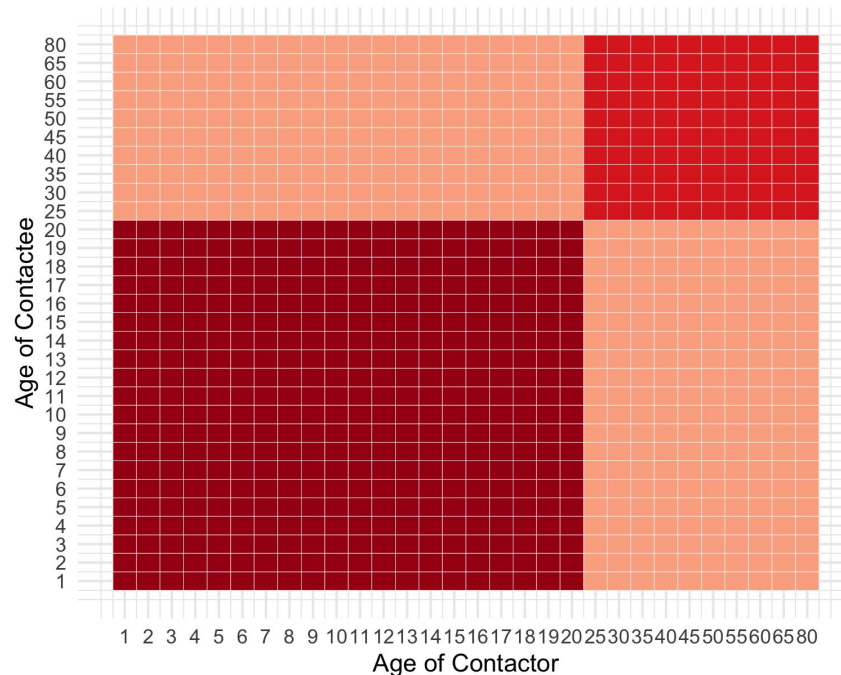
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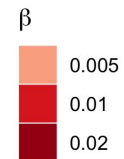
$$\begin{pmatrix} \beta_{\text{baby}, \text{baby}} & \beta_{\text{baby}, \text{teen}} \\ \beta_{\text{teen}, \text{baby}} & \beta_{\text{teen}, \text{teen}} \end{pmatrix}$$

2 age groups - WAIFW 2x2 matrix (4 β s)

diagonal (top left to bottom right) - assortative transmission



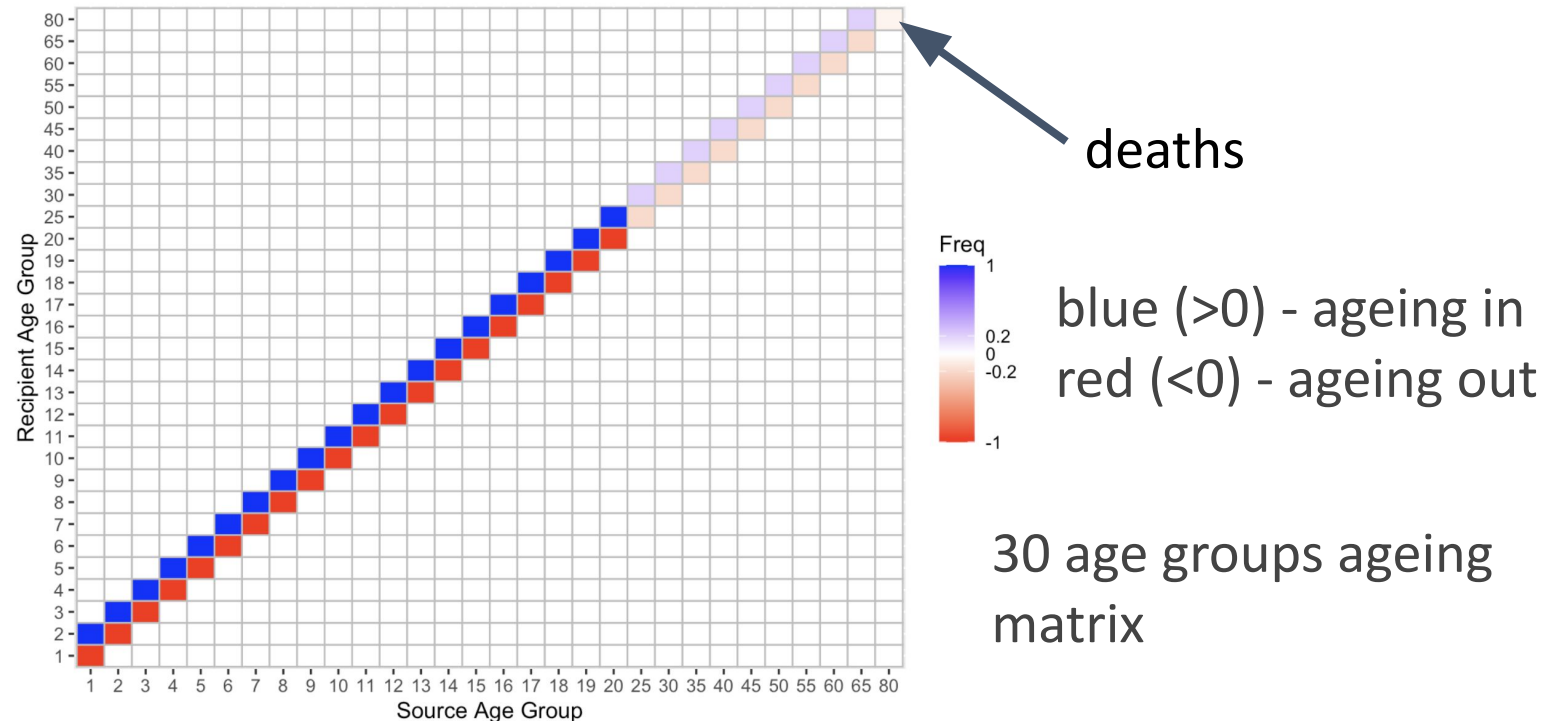
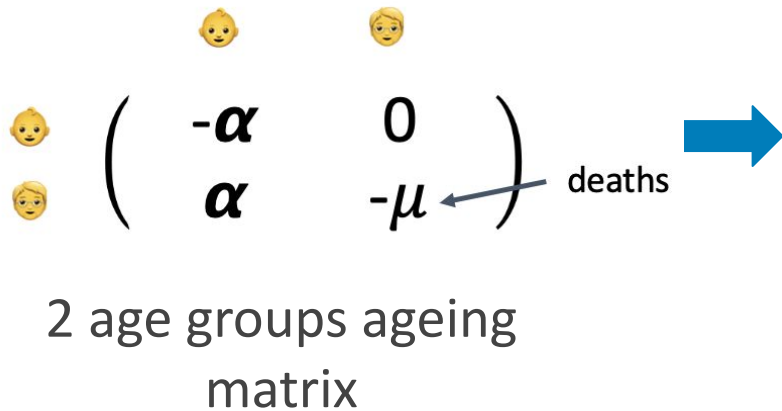
30 age groups -
WAIFW 30x30 matrix
(90 β s)



diagonal (bottom left
to top right) -
assortative
transmission

Modeling age heterogeneity

It is the **same as modeling heterogeneity with non-directional transitions** (i.e., age-specific parameters and a matrix of β or WAIFW to “talk” to each other), but **with the addition of individuals ageing**



A little demography

Characteristic	Stable Population	Stationary Population
Age Structure	Constant	Constant
Growth Rate	Constant ($\neq 0$)	Zero
Conditions	Constant births & deaths	births = deaths
Size Over Time	Changes (exponential)	Fixed

Examples of Age Structures:

